

Supply Chain Delivery Delay & Operational Risk Analysis

Comprehensive Business Intelligence & Machine Learning Project Report

Python | Pandas | Seaborn | Scikit-Learn | Imbalanced-Learn (SMOTE)

Target Application: E-Commerce Supply Chain, Logistics Optimization & Risk Mitigation

Dataset: DataCo Smart Supply Chain (180,519 raw rows, 53 columns; 172,765 orders after removing canceled shipments)

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1. Executive Summary

This report presents a comprehensive analysis of the delivery operations of a global e-commerce company managing end-to-end order fulfillment across multiple regions. The analysis covers **172,765 orders** spanning **January 2015 through January 2018**, focusing on identifying root causes of chronic late deliveries, quantifying their financial exposure, and establishing a data-driven framework for improvement.

The headline finding is stark: **54.71% of all orders are delivered late** (94,523 orders). Delayed orders carry **\$2.1M** of profit that is exposed to operational deterioration, concessions and churn, while total profit across profitable orders reached **\$7.5M**. The single biggest driver is shipping mode: **First Class is late on 100% of orders and Second Class on 79.8%**, while Standard Class is late on 39.8% and Same Day on 0%. A Random Forest model trained with SMOTE achieved **74% accuracy and 79% precision** on late orders.

Key Finding	Metric
Overall Late Delivery Rate	54.71%
On-Time Delivery Rate	45.29%
Total Orders Analyzed	172,765
Total Profit (profitable orders only)	\$7.5M
Profit Exposed on Delayed Orders	\$2.1M
First Class / Second Class Late Rate	100.0% / 79.8%
Predictive Model Accuracy (Random Forest)	74%
Predictive Model Precision, Late Class	79%

2. Business Context & Objectives

The company operates a global e-commerce platform selling sporting goods, fitness equipment, outdoor gear, footwear and apparel across multiple international regions. Actual shipping times frequently deviate from scheduled delivery windows, creating eroded customer trust, unpredictable order profitability, and an inability to make reliable delivery commitments at the point of purchase.

Analytical Objectives

- **Understand current delivery performance** across region, shipping mode, time and customer segment.
- **Quantify the financial impact** of delays on order profitability.
- **Identify the primary operational bottlenecks** driving late deliveries.
- **Build a predictive model** to flag high-risk orders before they are shipped.
- **Deliver actionable recommendations** to improve on-time delivery and profitability.

Data Preparation & Method

- **Cleaning:** removed free-text and identifier columns (names, email, password, street, zip codes, IDs), a fully empty column (Product Description), a single-value column (Product Status) and a duplicate of profit (Benefit per order). Canceled shipments were excluded, leaving 172,765 orders and 20 columns. No duplicate rows were found.

- **Delay definition:** Order Processing Time = shipping date minus order date (days). **Delay = Order Processing Time minus Scheduled Days.** An order is "late" when Delay is greater than 0.
- **Note on two late definitions:** the dataset flag Late_delivery_risk marks 98,977 orders (57.29%) as late, while the Delay-based definition marks 94,523 (54.71%). All business analysis uses the Delay-based 54.71%; the ML model predicts Late_delivery_risk.
- **Scheduled days are set by shipping mode:** Same Day = 0, First Class = 1, Second Class = 2, Standard Class = 4 days.

3. Key Performance Indicators (KPIs)

The following KPIs were computed directly from the cleaned dataset (172,765 non-canceled orders) to establish a performance baseline.

172,765 Total Orders	94,523 Late Deliveries	54.71% Late Delivery %	45.29% On-Time Delivery %
\$7.5M Total Profit (profitable orders)	\$2.1M Profit on Delayed Orders	3 days 90th Percentile Delay	\$22.03 Mean Profit / Order

KPI Insights & Diagnostics

• Late Delivery Rate at 54.71%	More than half of all orders arrive late. This is not an edge case. It is the default experience for the majority of customers.
• \$2.1M Profit on Delayed Orders	Delayed orders collectively generated \$2.1M in profit (net of any losses on those orders). This is profit exposed to concessions, churn and further deterioration. It is not money already lost.
• 90th Percentile Delay = 3 Days	Even the extreme cases are contained within 3 days of lateness, which suggests a systemic, process-driven problem rather than catastrophic one-off failures.
• \$7.5M vs Net Profit	The \$7.5M counts profitable orders only. Mean profit of \$22.03 across 172,765 orders implies a net profit of about \$3.8M once loss-making orders are included (derived figure).

4. Profitability Analysis

4.1 Profitability Distribution

Order-level profitability was classified into three tiers using Order Profit Per Order. While 80.7% of orders are profitable, the **18.7% loss-making share** is a meaningful drag on margin.

Profitability Tier	Order Count	Share (%)
Profit	139,354	80.7%
Loss	32,295	18.7%
Break-even	1,116	0.6%

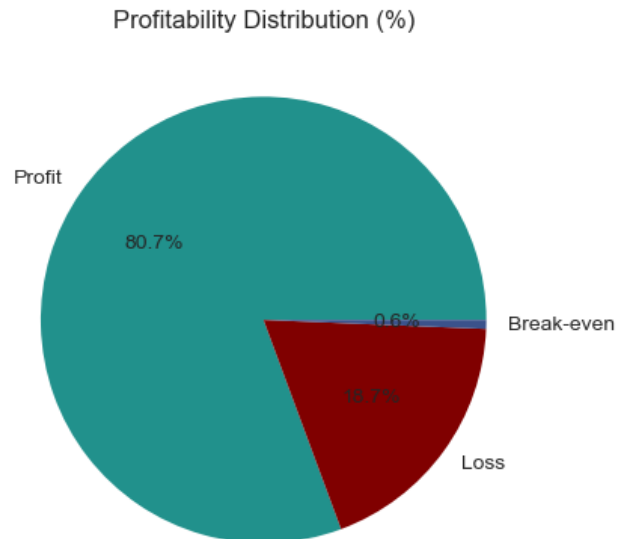


Figure 1: Profitability Distribution (%)

4.2 Delay Distribution & Profit vs. Delay Days

The delay distribution shows that **31.0% of all orders arrive exactly 1 day late**, the single largest cohort. Orders delayed by 1 to 4 days together account for **54.7%** of order volume, which maps directly to the overall late rate.

Delay (days)	Order Count	Order Share	Mean Profit	Total Profit
-2 (early)	20,873	12.1%	\$23.4	\$487.6K
-1 (early)	20,719	12.0%	\$21.6	\$447.6K
0 (on time)	36,650	21.2%	\$22.2	\$815.4K
+1 (late)	53,503	31.0%	\$22.3	\$1,194.9K
+2 (late)	27,551	15.9%	\$21.1	\$582.1K
+3 (late)	6,772	3.9%	\$20.0	\$135.7K
+4 (late)	6,697	3.9%	\$21.4	\$143.1K

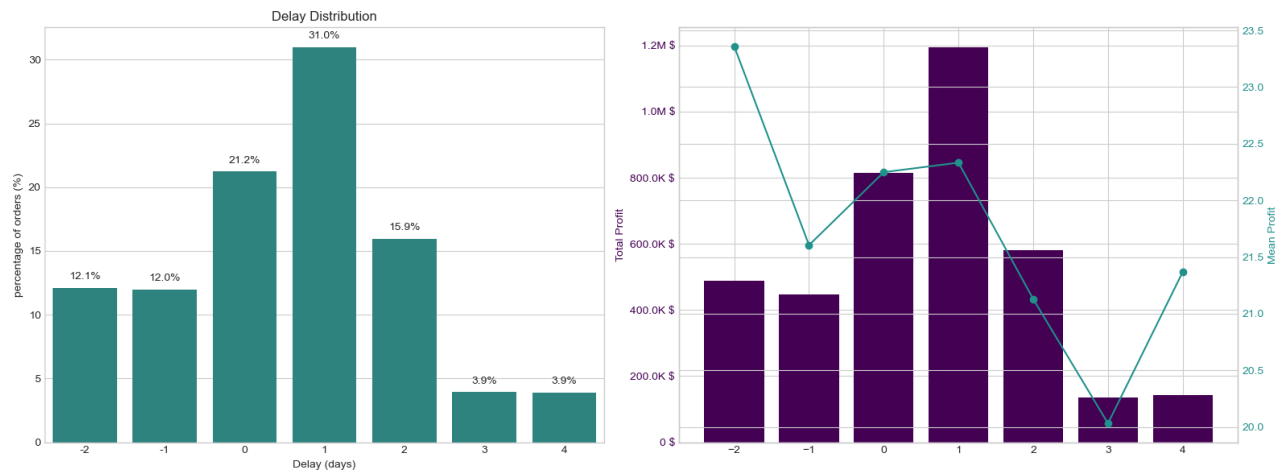


Figure 2: Delay Distribution (left) and Profit Analysis by Delay Days (right)

● Critical Observation

Mean profit per order stays at roughly \$20 to \$23 across all delay levels. The problem is driven by the **volume** of delayed orders, not by weaker unit economics on each order. Systemic throughput improvement is the highest-leverage intervention.

5. Bottleneck Detection

Delay percentage was computed across six operational dimensions (Order Region, Customer Segment, Shipping Mode, Order Status, Payment Type, Department Name). Charts show the top 10 values per dimension.

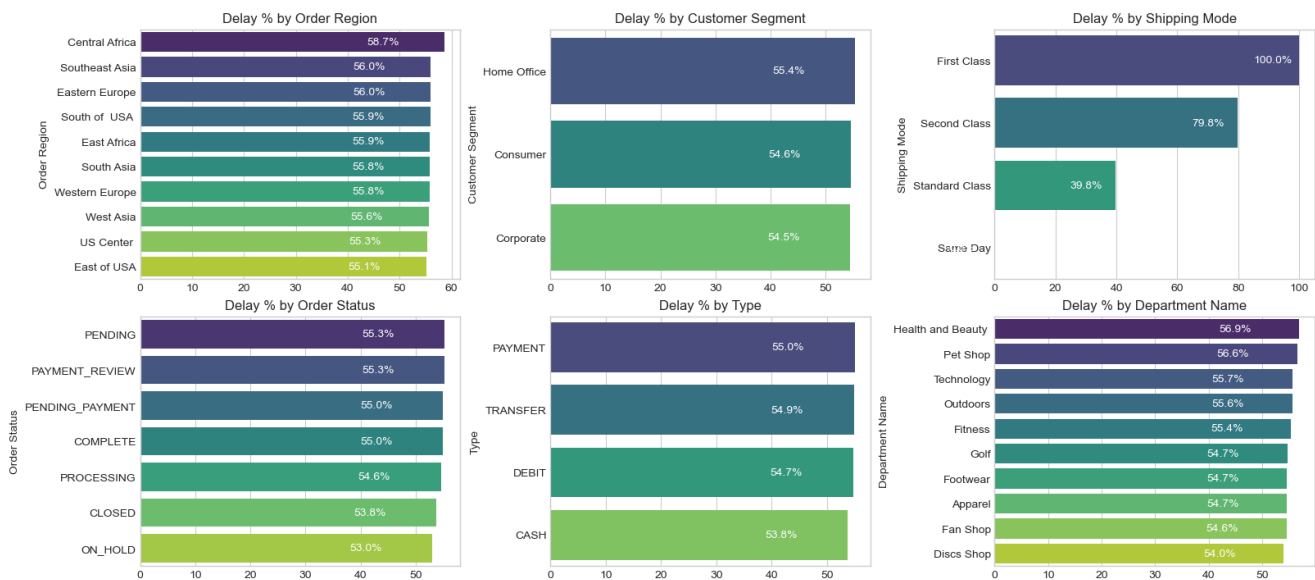


Figure 3: Bottleneck Detection, Delay % by Category

● Shipping Mode is the #1 Lever

First Class: **100.0%** late. Second Class: **79.8%**. Standard Class: **39.8%**. Same Day: **0%**. The faster the promised mode, the worse the delivery record.

● Regional Spread is Narrow

Central Africa leads at 58.7%, but the other top regions sit between 55.1% and 56.0%. This points to a company-wide systemic issue rather than one failing region.

● Customer Segments Nearly Identical	Home Office 55.4%, Consumer 54.6%, Corporate 54.5%. No segment receives better service.
● Order Status and Payment Type	Delay rates are almost flat: PENDING 55.3%, PAYMENT_REVIEW 55.3%, PENDING_PAYMENT 55.0%, COMPLETE 55.0%, ON_HOLD 53.0%. Payment type ranges only from 53.8% (Cash) to 55.0% (Payment). Payment status is not a global bottleneck.
● Departments	Health and Beauty (56.9%) and Pet Shop (56.6%) lead, marginally above the rest (54.0% to 55.7%). The gap is small.

6. Root Cause Analysis

Central Africa was selected for deep-dive as the highest-delay region (58.7%), and East Africa as a comparison hub. For each region the top 10 factor levels were ranked by delay percentage across Shipping Mode, Customer Segment, Department, Payment Type and Order Status.

Central Africa

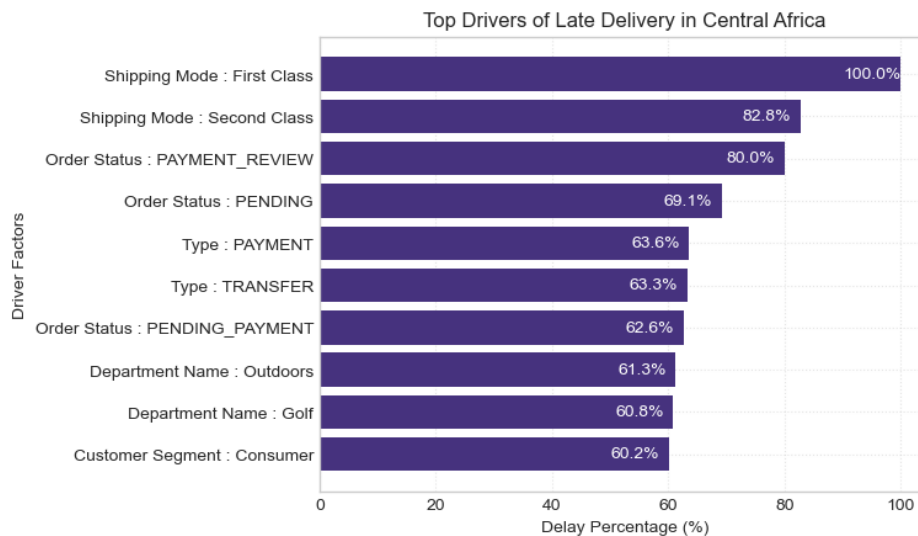


Figure 4: Top Drivers of Late Delivery in Central Africa

Rank	Driver Factor	Delay %
1	Shipping Mode : First Class	100.0%
2	Shipping Mode : Second Class	82.8%
3	Order Status : PAYMENT_REVIEW	80.0%
4	Order Status : PENDING	69.1%
5	Type : PAYMENT	63.6%
6	Type : TRANSFER	63.3%
7	Order Status : PENDING_PAYMENT	62.6%
8	Department : Outdoors	61.3%

Rank	Driver Factor	Delay %
9	Department : Golf	60.8%
10	Customer Segment : Consumer	60.2%

East Africa

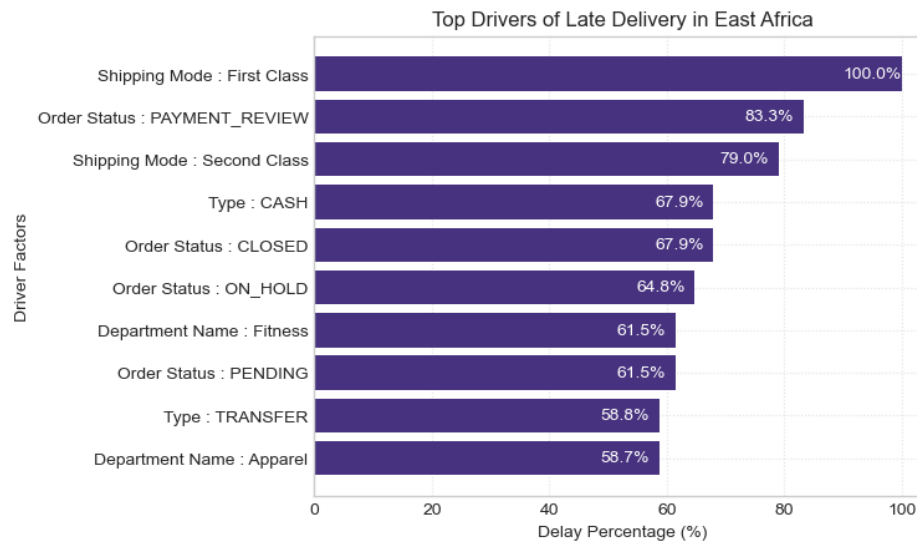


Figure 5: Top Drivers of Late Delivery in East Africa

Rank	Driver Factor	Delay %
1	Shipping Mode : First Class	100.0%
2	Order Status : PAYMENT_REVIEW	83.3%
3	Shipping Mode : Second Class	79.0%
4	Type : CASH	67.9%
5	Order Status : CLOSED	67.9%
6	Order Status : ON_HOLD	64.8%
7	Department : Fitness	61.5%
8	Order Status : PENDING	61.5%
9	Type : TRANSFER	58.8%
10	Department : Apparel	58.7%

• Key Takeaways

In both regions, **First and Second Class shipping** are the top failure points. **PAYMENT_REVIEW** orders are 80.0% (Central Africa) and 83.3% (East Africa) late regionally, even though they look average globally. In Central Africa, **Outdoors (61.3%)** and **Golf (60.8%)** are the weakest departments.

7. Time-Based Delay Patterns

Three time dimensions were analyzed: month of the year, day of the week and hour of the day, all based on order timestamp. Delay rates stay within a narrow band, so time is a secondary lever.

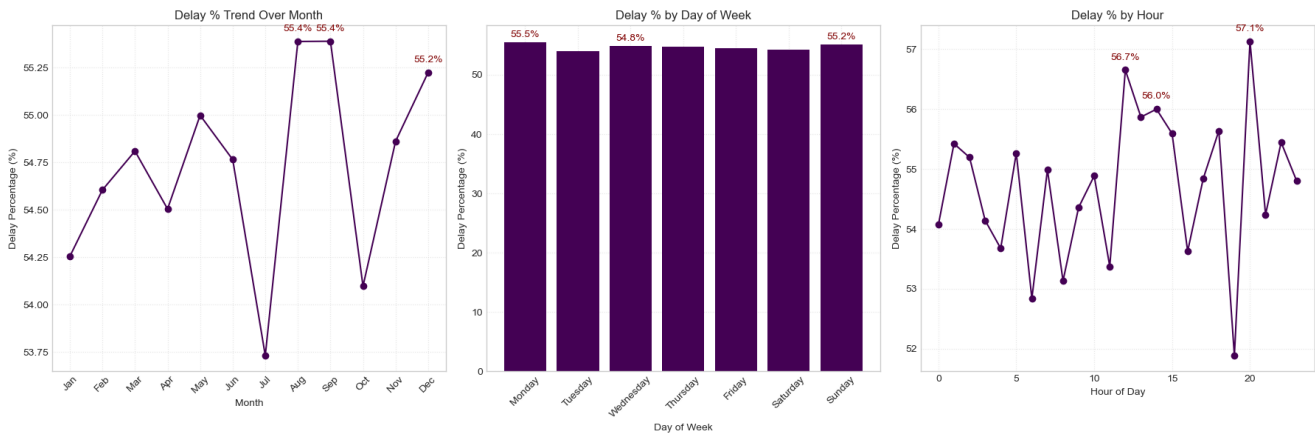


Figure 6: Delay % by Month, Day of Week and Hour of Day

Dimension	Peak Periods	Low Point	Observation
Month	Aug 55.4%, Sep 55.4%, Dec 55.2%	Jul about 53.7%	Total spread of about 1.7 points
Day of week	Mon 55.5%, Sun 55.2%, Wed 54.8%	Tuesday (lowest bar)	Spread about 1.6 points. Not a primary lever.
Hour of day	Hour 20 (8 PM) 57.1%, hour 12 56.7%, hour 14 56.0%	Hour 19 about 51.9%	Largest swing, about 5 points (51.9% to 57.1%)

• Interpretation

August, September and December are the highest-delay months and July is the lowest, which hints at seasonal load. Orders placed around 8 PM have the highest late rate. **Possible causes** (for example pickup cutoffs or shift changes) are hypotheses that this dataset cannot confirm and should be validated with operations data.

8. Machine Learning Model Results

A supervised classifier was built to predict whether an order will be late (Late_delivery_risk = 1).

Pipeline

- **Features (9 after encoding):** Days for shipment (scheduled), order month, order hour, plus frequency-encoded Type, Category Name, Customer Segment, Department Name, Order Region and Shipping Mode.
- **Frequency encoding:** each categorical level is replaced by its share of all orders, which avoids high-dimensional one-hot columns.
- **Split:** stratified 80/20 train-test split (random_state = 42), giving 138,212 training and 34,553 test records.
- **Class balancing:** SMOTE applied to the training set only.
- **Model:** Random Forest Classifier (default settings, random_state = 42).

Class Imbalance Handling (SMOTE)

Training Set	Class 0 (On-Time)	Class 1 (Late)	Total Records
Before SMOTE	59,030	79,182	138,212
After SMOTE	79,182	79,182	158,364

Random Forest: Test Evaluation (34,553 unseen records)

Class	Precision	Recall	F1-Score	Support
0 (On-Time)	0.68	0.73	0.70	14,758
1 (Late)	0.79	0.75	0.77	19,795
Macro average	0.73	0.74	0.73	34,553
Weighted average	0.74	0.74	0.74	34,553
Overall accuracy			0.74	34,553

• Model Viability	On late orders the model reaches 79% precision and 75% recall : about 4 in 5 orders it flags as late really are late, and it catches 3 in 4 of all late orders before dispatch.
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Limitations to Address Before Production

- **Shipping mode dominates the signal.** Scheduled days are fixed per mode, so those two features nearly encode the answer for First Class (always late) and Same Day (never late). Accuracy partly reflects this. Report feature importance and test the model on Standard and Second Class orders alone.
- **Encoding fitted on all data** before the train-test split. Fit frequency maps on the training set only to remove minor leakage.
- **No baseline or comparison model.** Add Logistic Regression and Gradient Boosting, a confusion matrix, and ROC-AUC.
- **No hyperparameter tuning** was performed.

9. Strategic Recommendations

Prioritized actions based on the findings above:

#	Priority	Finding	Action
1	CRITICAL	First Class 100% late, Second Class 79.8% late. Scheduled windows (1 and 2 days) are far tighter than delivered performance. Same Day has 0% late.	Audit carrier contracts and SLAs. Until lead times match promises, recalibrate promised delivery dates at checkout or pause First Class. Study why Same Day performs and apply those practices.
2	HIGH	Model gives 74% accuracy and 79% precision on late orders.	Validate the model without shipping-mode leakage, then expose it as a scoring service in the order system. High-risk orders trigger revised delivery dates and priority packing.

#	Priority	Finding	Action
3	HIGH	PAYMENT_REVIEW orders are 80.0% late in Central Africa and 83.3% in East Africa.	Set an escalation rule: orders in payment review beyond a fixed time limit go to prioritized manual verification. Review regional fraud-check workflows.
4	MEDIUM	Aug/Sep (55.4%) and Dec (55.2%) are peak delay months. Hour 20 is the worst hour (57.1%).	Secure extra carrier capacity before peak months and test an evening dispatch cutoff change. Validate causes with operations data first.
5	MEDIUM	Standard Class is late on 39.8% of orders versus 79.8% to 100% for faster modes.	Default checkout to the most reliable mode for regions and departments where premium modes consistently fail.
6	MEDIUM	Central Africa: Outdoors 61.3% and Golf 60.8% late. Health and Beauty (56.9%) and Pet Shop (56.6%) lead globally.	Run warehouse and carrier audits covering pick-to-pack time, stock availability and courier handoffs for these departments.
7	LOW	18.7% of orders lose money.	Review discounting on bulky and low-margin products. Check that shipping and return costs do not push thin-margin orders into loss.
8	LOW	Models degrade as routes and seasonality change.	Retrain quarterly. Add features such as warehouse backlog, weather and carrier tracking data to push recall above 80%.

10. Conclusion & Target Roadmap

This analysis shows clear visibility into delivery bottlenecks: **54.71%** of orders miss their scheduled window, putting **\$2.1M** of profit on delayed orders under pressure and eroding customer trust. The causes are identifiable and fixable:

- **Shipping mode promises are unrealistic.** First Class (100% late) and Second Class (79.8% late) are the primary breakdown.
- **Payment review holds** create a regional secondary delay, especially in Central and East Africa.
- **Seasonal and evening peaks** raise delay rates modestly and can be planned for.
- **The Random Forest model** is a useful early-warning prototype, provided the shipping-mode dependence is tested and reduced.

12-Month Operational Target Metrics (proposed planning goals)

Priority Area	Current Baseline	Target (12 Months)
Late Delivery Rate	54.71%	< 30.0%
First Class On-Time Rate	0.0%	> 80.0%
Second Class On-Time Rate	20.2%	> 60.0%
Model Accuracy (leakage-checked pipeline)	74.0%	> 80.0%
Loss-Making Orders Share	18.7%	< 12.0%
Profit Exposed on Delayed Orders	\$2.1M	Reduce late share to protect at least 40% (\$840K+)